A uniform contact certificate with independent source and testing parameters

The Bell slack is quadratic in the state coordinates. At a zero, every square in a positive decomposition must vanish. To retain this zero throughout the source family, we combine two square completions before estimating their singular source coefficients.

Write the source as $c = (d, \lambda, n, m, C, A, D, \alpha, \beta) \geq 0$ and put $D_u = 1 + u^2$, $D_v = 1 + v^2$. The canonical Bell slack $S_c = F_h/4$ is

$$\begin{aligned} D_u D_v S_c = {} & \lambda D_u (W + vX)^2 + n D_u D_v X^2 + m(vW - X + uvY - uZ)^2 \\ & + C D_v (uW - Y)^2 + A D_u D_v Y^2 + (D + d/2)(uW + uvX - Y - vZ)^2 \\ & + (\alpha + d/2) D_u (vY - Z)^2 + (\beta + d/2) D_v (X + uZ)^2 - \tfrac{d}{2} D_u D_v Z^2. \end{aligned} \tag{1}$$

The coupled state directions become explicit after the rotation

$$t = \frac{X + uZ}{D_u}, \quad r = \frac{Y - uW}{D_u}, \quad w = \frac{W + uY}{D_u}, \quad \zeta = \frac{Z - uX}{D_u}, \qquad (W, X, Y, Z) = (w - ur, t - u\zeta, r + uw, \zeta + ut). \tag{2}$$

This change is invertible for every real u and leaves all four state coordinates independent.

The source family in this note has $(d, \lambda, \beta) = (1, 0, 0)$. Its rational coefficients are given in Appendix A in terms of

$$\delta = 2^{-10}, \qquad 0 < B, T \leq \delta, \quad 0 < x = jB \leq \delta, \quad 0 < p < \min(j, 1), \quad 0 < H < 1, \quad n \geq 0. \tag{3}$$

The condition $n \geq 0$ is imposed on the displayed rational coefficient. Write $y = TB$. The ratios j and jT may tend to zero or infinity.

Theorem 1. *Every source in* (3) *satisfies* $S_c \geq 0$ *for every state and*

$$|u| \leq u_0 = 9/256, \qquad |v| \leq v_0 = 1/16.$$

On each required real sign chart the construction uses finitely many fixed testing functions and source-only positive Gram matrices. It includes $n = 0$ *literally and allows arbitrary nonnegative additions of the eight event coordinates. Source and testing parameters remain independent.*

Every native contact lies strictly inside this cap: its angles satisfy $|u| = \sqrt{x} \leq 1/32 < 9/256$ and $|v| = \sqrt{y} \leq 1/1024 < 1/16$. The certificate also covers testing angles unrelated to that contact.

The common source weight

$$\phi = \frac{p^2}{1 - p + p^2}$$

makes the two available completions compatible. The factor $1 - \phi$ cancels $1 - p$ in one denominator; ϕ controls the other denominator through a harmonic coefficient. The first-normal expression J_α below is kept whole while this allocation is made.

1 One weight for the two denominators

Put $y = TB$, $q_z = jT$, $\mu = m(1 - p)$ and define

$$\begin{aligned} & N = t - pvw, \quad M = -r - Hv\zeta, \quad b_{\mathrm{c}} = vw - u\zeta/j, \quad R = \zeta - uv\, w/B, \quad R_u = v(u^2 - x)w, \\ & h_0 = \mu + \tfrac{1}{2}, \quad M_e = \frac{n h_0 j^2}{n + h_0}, \quad \widehat{N} = \sqrt{n + h_0}\left(N + \frac{nj}{n + h_0} b_{\mathrm{c}}\right), \quad E_* = \sqrt{M_e}\, b_{\mathrm{c}}. \end{aligned} \tag{4}$$

Completing the square gives $h_0N^2+n(N+jb_{\rm c})^2=\widehat{N}^2+E_*^2$. The positive source masses f_c,q_c,r_c in Appendix A satisfy

$$\Sigma=M_e+f_c/y>0,\qquad p^2\le 32\Sigma,\qquad \theta=M_e/\Sigma\in[0,1]. \tag{5}$$

Appendix A proves this estimate without a bound on j. Consequently, with $D_p=1-p+p^2\ge 3/4$,

$$\frac{1-\phi}{1-p}=\frac{1}{D_p}\le\frac{4}{3},\qquad \frac{\phi}{\Sigma}\le\frac{128}{3},\qquad \frac{\phi}{j}\le\frac{4p}{3}\le\frac{4}{3}. \tag{6}$$

Every quantity in (6) depends only on the source.

To identify the corresponding squares, put

$$\begin{gathered}U=u/\sqrt{x},\quad V=v/\sqrt{y},\quad Z_*=\zeta/\sqrt{q_z},\quad Q_*=Uw-VZ_*,\quad B_*=Vw-UZ_*,\quad R_*=Z_*-UVw,\\ \Psi=(U-V)^2w^2+Z_*^2(1-UV)^2+2UV(w-Z_*)^2.\end{gathered} \tag{7}$$

Here $b_{\rm c}=\sqrt{y}\,B_*$ and $R=\sqrt{q_z}R_*$. The first-normal term and its exact decomposition are

$$J_\alpha=j\alpha b_{\rm c}=-\frac{\alpha}{B}R_u-\alpha uR. \tag{8}$$

Let $A_1=(\alpha/B)R_u$, $B_1=\alpha uR$. Since $(1+U^2)\Psi-B_*^2=R_*^2+U^2Q_*^2+U^2(w-UVZ_*)^2$,

$$\begin{aligned}&2(1-\phi)(\alpha/B)^2R_u^2+2(1-\phi)\alpha^2q_zu^2R_*^2+\phi j^2\alpha^2y(1+u^2/x)\Psi-J_\alpha^2\\ &\quad=(1-\phi)(A_1-B_1)^2+\phi j^2\alpha^2y[R_*^2+U^2Q_*^2+U^2(w-UVZ_*)^2].\end{aligned} \tag{9}$$

For $\eta=B+x$, the three coefficients on the left, relative respectively to R_u^2, $r_cR_*^2$ and $f_c\Psi$, are at most

$$2\eta^2,\qquad (64/3)\eta^2u^2,\qquad 8B\eta^2+(32/3)\eta^2u^2.$$

Thus keeping J_α whole adds only a small coefficient to R_u^2, even as $p\to 1$. The same ϕ gives bounds for R_u and the second-normal term

$$O=\sqrt{T}\,v(\alpha-BTCHu^2)R,\qquad H_1=pv^2b_{\rm c},\qquad H_2=\sqrt{T}\,puvb_{\rm c}. \tag{10}$$

Appendix B gives their exact identities and simultaneous bounds. Separating the source factors leaves fixed testing radicals.

2 Both normals in one positive matrix

Set

$$a=(n+h_0)^{-1/2},\quad b_0=\sqrt{\frac{n}{h_0(n+h_0)}},\quad \tau=\sqrt{TH(1-H)},\quad \rho_Q=T\sqrt{x/q_c},$$

$$\widehat{M}=M/\tau,\quad \widehat{Q}=(v\zeta-Tuw)/\rho_Q=-\sqrt{q_c}Q_*,\quad \widehat{R}=\sqrt{r_c}R_*,\qquad N=a\widehat{N}-b_0E_*.$$

In particular $a^2+b_0^2=1/h_0\le 2$. Choose the three explicit factors ψ_i of $f_c\Psi$ from (7) and use the thirteen rows

$$f=(\widehat{N},E_*,R_u,J_\alpha,H_1,\widehat{M},O,H_2,\widehat{Q},\widehat{R},\psi_1,\psi_2,\psi_3)^T.$$

The quadratic form f^TKf is specified by its diagonal

$$(1,\tfrac12,16,16,8,a_M,16,8,\tfrac12,\tfrac12,\tfrac14,\tfrac14,\tfrac14),\qquad a_M=\tau^2(C+D+\tfrac12+A), \tag{11}$$

and the following complete list of cross terms:

$$\begin{aligned}&2N(K_RR_u-J_\alpha+H_1/2)+2\sqrt{H(1-H)}\,\widehat{M}\{O+(\mu/p)H_2\}\\ &\quad+2uvN\{(K_c/j)\rho_Q\widehat{Q}+(\alpha+\tfrac12)\tau\widehat{M}\}+2\tau\rho_QC_q\,\widehat{M}\widehat{Q},\\ &K_c=p/2+j[H(\alpha+\tfrac12)-\tfrac12]+n(p-j),\\ &K_R=-\mu-(\alpha+\tfrac12)(1-p)+(T/j)K_c,\qquad C_q=A/T+Hv^2[C(1+u^2)+A+\alpha+\tfrac12].\end{aligned} \tag{12}$$

The source bounds give $|K_R| < 1$, $\mu/p \le 2/3$ and $1/8 < a_M < 3/4$. For the constant matrix excluding the constant $\widehat{M}\widehat{Q}$ entry, subtracting $I/16$ gives two Schur corrections at most

$$2(32/255 + 4/127) = 10168/32385 < 7/16, \qquad 4/255 + 16/1143 < 1/16.$$

Its remaining rows are diagonal. It therefore dominates $I/16$. The excluded entry has norm at most T, and the exact support is

$$K = K_{00} + uvK_{11} + v^2K_{02} + u^2v^2K_{22}, \qquad \|K_{11}\|, \|K_{02}\|, \|K_{22}\| \le 1. \tag{13}$$

The derivation is in Appendix A. Let Π_{11} be the coordinate projection onto $(\widehat{N}, E_*, \widehat{M}, \widehat{Q})$, and let $\Pi_{02} = \Pi_{22}$ project onto $(\widehat{M}, \widehat{Q})$. These projections contain the exact supports of the three nonconstant coefficient matrices. The fixed monomials $P_{ab} = (u/u_0)^a(v/v_0)^b$, for $(a,b) = (1,1), (0,2), (2,2)$, have budgets $9/4096, 1/256, 81/16777216$. The identity

$$K = K_{00} - \sum \rho_{ab}\Pi_{ab} + \sum_{a,b} \left[\frac{\rho_{ab}\Pi_{ab} + A_{ab}}{2}(1 + P_{ab}) + \frac{\rho_{ab}\Pi_{ab} - A_{ab}}{2}(1 - P_{ab}) \right], \quad A_{ab} = u_0^a v_0^b K_{ab}, \quad \rho_{ab} = u_0^a v_0^b$$

has positive source matrices and a constant reserve greater than $I/32$, because $1/16 - \delta - 9/4096 - 1/256 - 81/16777216 = 929711/16777216 > 1/32$.

3 Restoring the scalar squares

The complete parent identity, written out in Appendix C, is

$$D_v S_c = f^T K f + \mathcal{P} + \tfrac{3}{4} f_c \Psi + \tfrac{1}{2} q_c Q_*^2 + \tfrac{1}{2} M_e b_c^2 + \tfrac{1}{2} r_c R_*^2 - g_c \Psi - 2\kappa V^2 Q_*^2 - 2\kappa R_*^2 - \mathcal{T}, \tag{14}$$

where $\kappa = -k_c > 0$, $\mathcal{P}$ is a displayed sum of retained positive squares, and $\mathcal{T} = 16R_u^2 + 16J_\alpha^2 + 16O^2 + 8H_1^2 + 8H_2^2$. The identities in Appendix B replace all negative terms in (14) by positive remainders. Their total fractions b_i of $(f_c\Psi, q_c Q_*^2, M_e b_c^2, r_c R_*^2)$ satisfy

$$\|b_\Psi\|_* < 3/4, \qquad \|b_Q\|_* < 1/8, \qquad \|b_E\|_* < 1/8, \qquad \|b_R\|_* < 1/2. \tag{15}$$

The norm sums absolute coefficients weighted by $u_0^a v_0^b$. The core uses fractions $(1/4, 1/2, 1/2, 1/2)$, so each remaining scalar coefficient is strictly positive with a fixed coefficient reserve.

More explicitly, write $b_i = c_{i0} + \sum_\nu c_{i\nu} P_\nu$, where the nonconstant exponents lie in

$$(2,0), (4,0), (0,2), (2,2), (4,2), (6,2), (0,4), (2,4), (4,4).$$

Since these exponents are even, $0 \le P_\nu \le 1$ on the signed cap and P_ν is a polynomial square. With $c^\pm = \max(\pm c, 0)$,

$$1 - d_i - b_i = \underbrace{1 - d_i - c_{i0} - \sum_\nu c_{i\nu}^+}_{>0} + \sum_\nu c_{i\nu}^+ (1 - P_\nu) + \sum_\nu c_{i\nu}^- P_\nu.$$

This uses only fixed scalar gates. Expanding the finitely many displayed roots separates their source coefficients from their testing monomials. Appendix D records the resulting fixed lists.

All expressions are literal at $n = 0$: $b_0 = M_e = E_* = 0$ and $\Sigma = f_c/y > 0$. On $u, v \ge 0, w\zeta \le 0$, replace the three factors of $f_c\Psi$ by the five factors $\sqrt{f_c}(Uw, Vw, Z_*, UVZ_*, 2\sqrt{-UVwZ_*})$. Only the three uncoupled diagonal rows change to five rows with the same coefficient $1/4$; every matrix bound remains valid. Appendix C gives positive expansions of both radial forms on this state chart. The source-preserving angle gauges extend the two identities to all angle signs. All testing roots have their literal zero values on the axes; the independent core completion (46) also gives direct nonnegativity there. Adding nonnegative event coordinates in (1) preserves positivity. Complex states are covered by their independent real and imaginary parts.

For $|v| \ge 1/16$, the source coefficients C and D absorb the negative state square directly. Appendix E gives the corresponding ten-row matrix, valid for every Alice angle. Together the two certificates leave the R1 strip $|u| > 9/256$, $|v| < 1/16$, including Alice's projective endpoint, to be covered. An exact global lift also requires admission of the remaining canonical source families.

A Source coefficients and their uniform bounds

Every coefficient of the stationary source family is a rational function of B, T, j, p, H. We write these functions explicitly and bound the combinations used in the certificate. The symbol F in this appendix is a source polynomial, distinct from the homogeneous slack F_h. Set $x = jB$, $y = TB$, $q_z = jT$, and define

$$\begin{aligned}
K &= j + 1 - p, \quad E = 1 + B(HT + p - 1), \quad W_0 = (1 - H)j + 1 - p,\\
P &= j + 1 + B((j+1)^2 - p) + TB[Hj + j + 1 + B(Hj(2j + 2 - p) + p(2j + 1 - p))]\\
&\quad + HjT^2B^2(1 + Bp + BHj),\\
L &= K + T(W_0 - H) + BT[pK + Hj(1 - p)] + BHT^2(W_0 - 1),\\
\mathsf{N} &= p + B(-j^2 + 3jp - j - p^2 + p) + BT(Hj^2 + Hj - j^2 + jp - j + p)\\
&\quad + B^2T(2Hj^2p - Hj^2 - Hjp^2 + 2Hjp - 2j^2p + 3jp^2 - jp - p^3 + p^2) + B^2HT^2j(Hj - j + p)(1 + Bp),\\
F_1 &= -2BHjp + BHj + 2Bjp - 2Bp^2 + 2Bp - Hj - Hp - H + j + 1 - p^2/j + p/j,\\
F_2 &= -B^2H^2jp + B^2Hjp - B^2Hp^2 + B^2Hp - BH^2j - BH^2p + BHj - BHp^2/j + BHp/j,\\
F &= K + TF_1 + T^2F_2,\\
\mathsf{K} &= pW_0 + B(1 - p)[j(H(p + 1) - p) - (1 - p)(p - H)]\\
&\quad + TBH[(p + 1)(1 - H)j + (1 - p)(BHj + p + 1 - H)] + T^2B^2H^2j(1 - H).
\end{aligned} \tag{16}$$

The six native weights and the scalar masses are

$$\begin{aligned}
\mu &= \frac{jp(1+y)(1+x)(1+Hy)}{2P}, & m &= \frac{\mu}{1-p},\\
n &= \frac{\mathsf{N}}{2(j-p)P}, & C &= \frac{L}{2HTP},\\
A &= \frac{T(1+x)(1+Hy)(K+Hjy)}{2(1-HT)P},\\
D &= \frac{(1+Hy)(K+Hjy)[1-y((1-H)j-p)]}{2T(1-H)P}, & \alpha &= \frac{B^2(K+Hjy)L}{2EP},\\
f_c &= \frac{Tx^2F}{2P}, & q_c &= Ax - f_c,\\
r_c &= \frac{B^2jT(1-p)(1+x)(1+Hy)L}{2EP},\\
g_c &= \frac{HT^2j^2B^3L}{2P}, & k_c &= -\frac{T^2B^2j(1+x)\mathsf{K}}{2EP},\\
d_c &= H^2(A + C + \alpha + \tfrac{1}{2})y^2jT - k_c.
\end{aligned} \tag{17}$$

The condition $n \geq 0$ means $\mathsf{N} \geq 0$. Every denominator in this dictionary is positive on (3). In particular $1 - HT > 0$, $1 - y[(1 - H)j - p] \geq 1 - Tx > 0$ and $E \geq 1 - \delta > 0$.

Put $J = j + 1$, $K = J - p$, and

$$f_0 = 1 - \delta - \delta^3, \quad \ell_0 = 1 + \delta + \delta^2 + \delta^3, \quad R_0 = 1 + 2\delta + 2\delta^2 + 4\delta^3 + \delta^4(1 + 2\delta).$$

Because $p < \min(j, 1)$, $J/2 \leq K < J$ and $K > 1$. Termwise bounds in the displayed polynomials give

$$J \leq P \leq R_0J, \qquad 1 - \delta \leq E \leq 1 + \delta^2, \qquad f_0K \leq L \leq \ell_0K, \qquad f_0K \leq F \leq 2K. \tag{18}$$

Here $P \geq J$ follows from its positive terms; in the upper bound use $BJ = B + x \leq 2\delta$ to bound its successive corrections, divided by J, by $2\delta, 2\delta^2, 4\delta^3, \delta^4(1 + 2\delta)$. For L, use $W_0 = K - Hj$ to obtain

$$L \geq K - T - BT^2 \geq f_0K, \qquad L \leq K[1 + T + BT + BT^2] \leq \ell_0K.$$

For F, the exact regrouping

$$\begin{aligned}
F_1 &= 2Bjp(1 - H) + BHj + 2Bp(1 - p) + (1 - H)J - Hp + p(1 - p)/j,\\
F_2 &= B^2Hjp(1 - H) + B^2Hp(1 - p) + BHj(1 - H) - BH^2p + BHp(1 - p)/j
\end{aligned}$$

gives $F \ge K - Tp - BT^2 p \ge f_0 K$. For the upper bound, $F_1 \le (3+5\delta)K$ and $F_2 \le (2\delta + 2\delta^2)K$, which yield $F < 2K$ at $\delta = 2^{-10}$.

Write

$$G_q = \frac{(1+x)(1+Hy)(K+Hjy)}{1-HT} - xF, \qquad q_c = \frac{TxG_q}{2P}.$$

The first term in G_q is at least K, so $G_q \ge (1-2\delta)K$. The dictionary consequently gives

$$f_c/y > j^2 B/5, \quad f_c \ge Tx^2/8, \quad q_c \ge Tx/8, \quad r_c \ge TjB^2(1-p)/8, \quad \alpha \le B(B+x), \quad c_0 := BTCH < B. \tag{19}$$

For the first four inequalities use, respectively,

$$\frac{f_0}{4R_0} > \frac{1}{5}, \qquad \frac{f_0}{4R_0} > \frac{1}{8}, \qquad \frac{1-2\delta}{4R_0} > \frac{1}{8}, \qquad \frac{f_0}{4(1+\delta^2)R_0} > \frac{1}{8}.$$

For α, bound $(K+Hjy)/P \le 1+\delta^2$ and $L \le \ell_0 J$:

$$\alpha \le B^2 J \frac{(1+\delta^2)\ell_0}{2(1-\delta)} \le B(B+x).$$

The last scalar factor is less than one. Also $c_0 = BL/(2P) < B$. Directly,

$$\mu/p \le (1+\delta)(1+\delta^2)^2/2 < 2/3, \qquad A/T \le \frac{(1+\delta)(1+\delta^2)^2}{2(1-\delta)} < 2/3. \tag{20}$$

The harmonic bound

Put $\rho = p/j < 1$. Bounding the negative terms of N/j below and discarding only nonnegative terms gives

$$\begin{aligned} \mathsf{N}/j \ge \rho - BD_N, \qquad & D_N = 1 + 2j + T(1+j) + BT(2j+4j^2) + BT^2 j(1+Bj), \\ & D_N/J \le 2 + \delta + 4\delta^2 + \delta^3(1+\delta) < 9/4. \end{aligned} \tag{21}$$

Since $\mathsf{N}/j = 2(1-\rho)Pn$ and $\rho J = p + \rho < 2$,

$$\rho^2 \le \frac{2\rho}{J} \le 4(P/J)n + 2B(D_N/J) \le 5n + \tfrac{9}{2}B.$$

If $n \le h_0$, $M_e \ge nj^2/2$, and hence

$$p^2 \le 5nj^2 + \tfrac{9}{2}Bj^2 \le 10M_e + \tfrac{45}{2} f_c/y \le \tfrac{45}{2}\Sigma.$$

If $n \ge h_0$, then $M_e \ge h_0 j^2/2 \ge j^2/4$ and $p^2 < j^2 \le 4M_e$. This proves (5). At $n = 0$ the positive term f_c/y is unchanged.

Signed coefficients and the normal matrix

The exact positive expression

$$\begin{aligned} \mathsf{K} - pW_0 + Bp(1-p)J = {} & B(1-p)[Hj(p+1) + H(1-p) + p^2] \\ & + TBH[(p+1)(1-H)j + (1-p)(BHj + p + 1 - H)] \\ & + T^2 B^2 H^2 j(1-H) \end{aligned}$$

shows $\mathsf{K} > 0$, since $BJ < 1$ and $W_0 \ge 1 - p$. For an upper bound, its four displayed lines contribute at most $p, B, B\delta(2+\delta), B\delta^3$ after division by J. Thus $\mathsf{K}/J \le p + 3B$ and $\kappa = -k_c > 0$; the exact ratios imply

$$\phi\kappa/f_c \le 3T, \quad \phi\kappa/(\Sigma y x) \le 24T, \quad \kappa/(q_c y) \le 3, \quad (1-\phi)\kappa/r_c \le 3T, \quad g_c/f_c = HyL/F \le 2y. \tag{22}$$

For example the four rational constants proving the first four bounds are bounded, respectively, by

$$\frac{8(1+\delta)(1+3\delta)}{3(1-\delta)f_0} < 3, \quad \frac{64(1+\delta)(1+3\delta)}{3(1-\delta)} < 24, \quad \frac{(1+\delta)(2+6\delta)}{(1-\delta)(1-2\delta)} < 3, \quad \frac{4(2+6\delta)}{3f_0} < 3.$$

They use (6), $K \ge J/2$ and the exact source ratios.

For the first normal, bounding the positive terms in N/j above and discarding its nonpositive terms gives

$$\mathsf{N}/j \le 1 + 4\delta + 2\delta^2 + 6\delta^3 + \delta^4(1+\delta) < 2.$$

In this bound use $Hj + H - j - 1 \le 0$ and $2Hjp - 2jp \le 0$ before bounding the other terms. As $P \ge J$, it follows that $0 \le n(1-p/j) = \mathsf{N}/(2jP) < 1$. Therefore $|K_c/j| \le 2$, and

$$|K_R| \le \mu + (\alpha + \tfrac{1}{2})(1-p) + 2T \le \tfrac{2}{3} + 2\delta^2 + 2\delta < 1.$$

For the second normal, the complementary H and $1-H$ factors in C, D must be retained together:

$$a_M = TH(1-H)(C + D + \tfrac{1}{2} + A), \qquad \frac{f_0}{4R_0} \le a_M \le \frac{\max\{\ell_0, (1+\delta^2)^3\}}{2} + \frac{\delta}{8} + \frac{\delta^2}{6}.$$

The endpoints lie strictly between $1/8$ and $3/4$. Indeed $THC = L/(2P)$, while $T(1-H)D = (1+Hy)(K+Hjy)[1 - y((1-H)j - p)]/(2P)$. Their lower bounds are $f_0/(4R_0)$ and $(1-\delta^2)/(4R_0)$, and their upper bounds are $\ell_0/2$ and $(1+\delta^2)^3/2$. The terms $\tau^2/2$ and $\tau^2 A$ contribute at most $\delta/8$ and $\delta^2/6$.

Finally $\rho_Q \le \sqrt{8T}$ and $\tau \le \sqrt{T}/2$. Reading the three cross coefficients in (12) gives

$$\|K_{11}\|^2 \le 578T/9 < 1,$$
$$\|K_{02}\| \le \sqrt{2}[\ell_0/2 + \delta(2\delta/3 + 2\delta^2 + \tfrac{1}{2})] < 1, \qquad \|K_{22}\| \le \sqrt{2}\,\ell_0/2 < 1.$$

The constant $\widehat{M}\widehat{Q}$ entry is at most $\tau\rho_Q A/T \le 2\sqrt{2}\,T/3 < T$. This proves every matrix bound used in the main argument.

B Exact identities and simultaneous coefficient bounds

In this appendix $s = u^2$ and $\eta = B + x$. We compare positive multiples of the four forms

$$(f_c\Psi,\ q_cQ_*^2,\ M_e b_{\mathrm{c}}^2,\ r_cR_*^2).$$

The fractions (b_Ψ, b_Q, b_E, b_R) bound a row when the sum of these four weighted forms minus its square is an explicitly displayed sum of squares.

The elementary identities needed below are

$$\begin{aligned}\mathcal{A} &:= (1+U^2)\Psi - R_*^2 = B_*^2 + U^2Q_*^2 + U^2(w - UVZ_*)^2,\\ \mathcal{E} &:= (1+V^2)\Psi - V^2B_*^2 = Q_*^2 + R_*^2 + V^2(w - UVZ_*)^2.\end{aligned} \tag{23}$$

There is also the opposite form

$$\begin{aligned}\mathcal{B} := 2(\Psi + V^2Q_*^2 + U^2B_*^2) - R_*^2 &= (E_0+T_0)^2 + 2(U-V)^2w^2 + 4UV(w-Z_*)^2\\ &\quad + 2J_0^2 + 4UV(U-V)^2Z_*^2,\\ E_0 = Z_*(1-UV), \quad T_0 = UV(w-Z_*), &\quad J_0 = T_0 - (U-V)^2Z_*.\end{aligned} \tag{24}$$

All are sums of squares for $U, V \ge 0$.

A common bound for the opposite row

For a nonnegative polynomial multiplier a_0, define the fractions

$$\begin{aligned}\mathfrak{b}_\Psi(a_0) &= a_0\phi\left[\frac{1+\theta}{f_c} + \frac{s}{\Sigma yx}\right], & \mathfrak{b}_Q(a_0) &= \frac{2a_0\phi\theta v^2}{q_cy},\\ \mathfrak{b}_E(a_0) &= \frac{2a_0\phi s}{\Sigma yx}, & \mathfrak{b}_R(a_0) &= \frac{a_0(1-\phi)}{r_c}.\end{aligned} \tag{25}$$

Since $f_c/(\Sigma y) = 1-\theta$ and $M_e/\Sigma = \theta$, exactly

$$\begin{aligned}f_c\mathfrak{b}_\Psi(a_0)\Psi + q_c\mathfrak{b}_Q(a_0)Q_*^2 + M_e\mathfrak{b}_E(a_0)b_{\mathrm{c}}^2 + r_c\mathfrak{b}_R(a_0)R_*^2 - a_0R_*^2\\ = a_0\phi[(1-\theta)\mathcal{A} + \theta\mathcal{B}].\end{aligned} \tag{26}$$

The direct fraction on R_*^2 is weighted by $1-\phi$ before r_c^{-1} is estimated.

For $O=\sqrt{j}\,Tv(\alpha-c_0 s)R_*$, put

$$\bar a = 2jT^2v^2(\alpha^2+c_0^2s^2), \qquad c_0 = BTCH.$$

There is the exact additional square

$$\bar a R_*^2 - O^2 = jT^2v^2(\alpha+c_0 s)^2R_*^2.$$

Therefore (26) with $a_0=\bar a$ bounds O^2. Using (6) and (19), its coefficientwise absolute majorants are

$$\begin{aligned}
\mathfrak{b}_\Psi(\bar a) &\le_* (128/3)Tv^2(1+2s)(\eta^2+s^2),\\
\mathfrak{b}_Q(\bar a) &\le_* 32v^4(\eta^2+s^2),\\
\mathfrak{b}_E(\bar a) &\le_* (512/3)Tsv^2(\eta^2+s^2),\\
\mathfrak{b}_R(\bar a) &\le_* (64/3)Tv^2(\eta^2+s^2).
\end{aligned} \tag{27}$$

Here $\le_*$ compares absolute coefficients as polynomials in s, v^2.

The entire signed term is bounded by taking $a_0=2\kappa$ in (26) and adding $2\kappa V^2Q_*^2$. Indeed

$$2\kappa V^2Q_*^2 + 2\kappa R_*^2 - \kappa(VQ_*+R_*)^2 = \kappa(VQ_*-R_*)^2.$$

The four fractions in this bound have majorants

$$(12T+48Ts, \quad 18v^2, \quad 96Ts, \quad 6T). \tag{28}$$

To see this directly, use the four bounds in (22) in (25); for the Q fraction use $[4\phi\theta+2]\kappa/(q_c y)\le 18$.

The radial row

The positive polynomial $\mathcal{W}$ and its permuted version $\mathcal{W}^\pi$ are given in Appendix C. Define

$$\mathcal{R}_A = \mathcal{W} + 4U^2\mathcal{A}, \qquad \mathcal{R}_B = 2U^2\mathcal{W}^\pi + [UZ_*(U^2-1)-(U^2-1)B_*]^2.$$

The expansions in Appendix C show that they are sums of squares on both state charts. Let

$$\begin{aligned}
r_\Psi &= (1-\phi)\frac{4x^2y}{f_c} + \phi\left[\frac{8\theta xys}{f_c} + \frac{4(x^2+xs+s^2)}{\Sigma}\right],\\
r_Q &= (1-\phi)\frac{4xys}{q_c} + \phi\frac{8\theta ys^2+4(1-\theta)xys}{q_c},\\
r_E &= \phi\frac{8s^2+2(s-x)^2}{\Sigma}, \qquad r_R = (1-\phi)\frac{4xys}{r_c}.
\end{aligned} \tag{29}$$

Then the exact identity is

$$f_c r_\Psi \Psi + q_c r_Q Q_*^2 + M_e r_E b_c^2 + r_c r_R R_*^2 - R_u^2 = x^2y[(1-\phi)\mathcal{W} + \phi(1-\theta)\mathcal{R}_A + \phi\theta\mathcal{R}_B]. \tag{30}$$

Expansion using (23) and $Vw(U^2-1) = UZ_*(U^2-1)+(U^2-1)B_*$ proves the identity. The two fractions containing the strongest source dependence are best bounded together with their coefficients. Put $J=j+1$, $K=j+1-p$ and use the positive constants f_0, R_0 from Appendix A. The source bounds give $P/F \le (R_0/f_0)J/K$. For $q=j-p>0$, direct expansion gives

$$\begin{aligned}
4jKD_p - J[j(1-p)+2p^2] &= (4p^2-3p+3)q^2\\
&\quad + (p+1)(4p^2-4p+3)q + 3p(1-p)^2 \ge 0.
\end{aligned} \tag{31}$$

Both quadratic coefficients are positive for real p. Dividing by $jKD_p>0$ therefore proves

$$\frac{J}{K}\left(1-\phi+\frac{2\phi}{j}\right) \le 4.$$

This retains the relation between the two source factors; bounding them separately would give a smaller testing region.

Substitution of f_c and r_c in (29) yields

$$r_\Psi = \frac{8P}{F}\left[B(1-\phi)+\frac{2\phi\theta s}{j}\right]+\frac{4\phi}{\Sigma}(x^2+xs+s^2),$$
$$r_R = \frac{8PEs}{D_p(1+x)(1+Hy)L}. \tag{32}$$

The coefficients of these polynomials in s are nonnegative. For any squared Alice cap $S \ge \delta$, their weighted coefficient norms obey

$$\|r_\Psi\|_* \le 32(R_0/f_0)S + (512/3)(\delta^2+\delta S+S^2),$$
$$\|r_R\|_* \le (64/3)R_0(1+\delta^2)S/f_0. \tag{33}$$

For the first inequality, use $B \le \delta \le S$, $\theta \le 1$ and (31). For the second, use $E \le 1+\delta^2$, $L \ge f_0K$, $J/K \le 2$ and $D_p \ge 3/4$. The condition $S \ge \delta$ concerns the fixed cap, not individual testing points, so the inequalities include $u = 0$.

The other two radial fractions have coefficientwise majorants

$$r_Q \le_* 64Bs + (256/3)s^2, \qquad r_E \le_* (1280/3)(\eta+s)^2. \tag{34}$$

Indeed $xy/q_c \le 8B$ and $y/q_c \le 8/j$; then use $\phi/j \le 4/3$ and $\phi/\Sigma \le 128/3$. For $(s-x)^2$, the absolute-coefficient majorant is $(s+x)^2$.

The bound (9) for $16J_\alpha^2$ includes $32(1-\phi)(\alpha/B)^2R_u^2$. Thus the effective radial multiplier is the source-only number

$$L_R = 16 + 32(1-\phi)(\alpha/B)^2 \le 16 + 32\eta^2 < 17. \tag{35}$$

Its remaining $f_c\Psi$ and $r_cR_*^2$ fractions are

$$j_\Psi = \frac{16\phi j^2\alpha^2 y(1+s/x)}{f_c}, \qquad j_R = \frac{32(1-\phi)\alpha^2 q_z s}{r_c}. \tag{36}$$

Their majorants are $16[8B\eta^2 + (32/3)\eta^2 s]$ and $16(64/3)\eta^2 s$, respectively.

The two remaining helpers

Put $\gamma = p^2/\Sigma \le 32$. From (23),

$$\gamma M_e v^4 b_c^2 + \gamma f_c(yv^2+v^4)\Psi - H_1^2 = \gamma f_c y v^2 \mathcal{E},$$
$$\gamma T M_e s v^2 b_c^2 + \gamma T f_c s(y+v^2)\Psi - H_2^2 = \gamma T f_c y s \mathcal{E}. \tag{37}$$

Both are polynomial identities, including on the testing axes. The total fractions for $8H_1^2 + 8H_2^2$ are therefore

$$h_\Psi = 8\gamma[yv^2+v^4+Ts(y+v^2)], \qquad h_E = 8\gamma[v^4+Tsv^2].$$

All scalar coefficients at once

There are two equivalent ways to retain the signed square. The complete-parent decomposition (14) has already used its exact remainder $\kappa(VQ_* - R_*)^2$, leaving $2\kappa V^2Q_*^2 + 2\kappa R_*^2$ to be bounded. Define its fractions by (25) with $a_0 = 2\kappa$, plus $2\kappa V^2/q_c$ in the Q component. Then the exact total losses are

$$\begin{pmatrix} b_\Psi \\ b_Q \\ b_E \\ b_R \end{pmatrix} = L_R \begin{pmatrix} r_\Psi \\ r_Q \\ r_E \\ r_R \end{pmatrix} + \begin{pmatrix} j_\Psi \\ 0 \\ 0 \\ j_R \end{pmatrix} + 16 \begin{pmatrix} \mathfrak{b}_\Psi(\bar{a}) \\ \mathfrak{b}_Q(\bar{a}) \\ \mathfrak{b}_E(\bar{a}) \\ \mathfrak{b}_R(\bar{a}) \end{pmatrix} + \begin{pmatrix} \mathfrak{b}_\Psi(2\kappa) \\ \mathfrak{b}_Q(2\kappa) + 2\kappa V^2/q_c \\ \mathfrak{b}_E(2\kappa) \\ \mathfrak{b}_R(2\kappa) \end{pmatrix} + \begin{pmatrix} h_\Psi + g_c/f_c \\ 0 \\ h_E \\ 0 \end{pmatrix}. \tag{38}$$

Every replacement in this equality has the exact positive remainder displayed above; the last g_c/f_c bounds $g_c\Psi$ directly. The radial term from (9) is included in L_R.

For explicit rational upper bounds, set

$$S = u_0^2 = 81/65536, \qquad V_0 = v_0^2 = 1/256, \qquad e = 2\delta, \qquad D_0 = e^2 + S^2.$$

Apply the nonnegative coefficient majorants at $s = S, v^2 = V_0$. The four resulting sums are bounded by

$$\begin{aligned}
A_\Psi &= 17[32(R_0/f_0)S + (512/3)(\delta^2 + \delta S + S^2)] + 16[8\delta e^2 + 32e^2 S/3] \\
&\quad + (2048/3)\delta V_0(1+2S)D_0 + 12\delta + 48\delta S \\
&\quad + 256(\delta^2 V_0 + V_0^2 + \delta^3 S + \delta S V_0) + 2(\delta^2 + V_0) < 3/4, \\
A_Q &= 17(64\delta S + 256S^2/3) + 512V_0^2 D_0 + 18V_0 < 1/8, \\
A_E &= (21760/3)(e+S)^2 + (8192/3)\delta S V_0 D_0 + 96\delta S + 256(V_0^2 + \delta S V_0) < 1/8, \\
A_R &= (1088/3)R_0(1+\delta^2)S/f_0 + (1024/3)e^2 S + (1024/3)\delta V_0 D_0 + 6\delta < 1/2.
\end{aligned} \tag{39}$$

These are strict rational comparisons at the displayed cap endpoints. The last term in A_Ψ is a conservative bound on $g_c/f_c \le 2y$. Thus (39) proves (15) as coefficient norm bounds, not only as pointwise inequalities. In particular it proves positivity of the fixed scalar reserves in the main text.

C The complete parent and its positive remainders

For a direct expansion of (1), put

$$q = v\zeta - Tuw = -T\sqrt{x}\,Q_*, \qquad m_p = \frac{\mu p}{1-p},$$

and use the two independent normals N, M in (4). Let

$$\begin{aligned}
F_s &= q_c Q_*^2 + nj^2 y B_*^2 + r_c R_*^2 + d_c V^2 Q_*^2 + (np^2y^2 + \tfrac{1}{2}p^2y^2 - g_c)V^2 B_*^2 \\
&\quad + f_c\Psi + g_c V^2\Psi + k_c Z_*^2 (V^2-1)^2, \\
L_N &= njb_{\mathrm{c}} + npv^2 b_{\mathrm{c}} - j\alpha b_{\mathrm{c}} + \tfrac{1}{2}pv^2 b_{\mathrm{c}} + K_R R_u + (K_c/j)uvq, \\
L_M &= C_q q + \mu uv b_{\mathrm{c}} + v(\alpha - BTCHu^2)R, \\
H_N &= h_0(1+u^2) + \tfrac{1}{2}v^2 + \alpha u^2 + n(1+v^2), \\
H_M &= (1+u^2)[C(1+v^2) + D + \tfrac{1}{2}] + A(1+v^2) + (\alpha + \tfrac{1}{2})v^2.
\end{aligned} \tag{40}$$

The full parent is the polynomial identity

$$D_v S_c = F_s + 2L_N N + 2L_M M + H_N N^2 + H_M M^2 + 2(\alpha + \tfrac{1}{2})uvNM + m_p(1+u^2)N^2. \tag{41}$$

It follows by inserting (2) into (1) and using (16)–(17). The four scalar cancellations needed in this expansion are, with $K_d = CH - (D + \frac{1}{2})(1-H)$,

$$\begin{aligned}
0 &= T(K_d + AH + \tfrac{1}{2}) - A, \\
0 &= jK_d + \mu + xTCH, \\
0 &= jK_d + xTCH + T[CH + AH + (\alpha + \tfrac{1}{2})H] - A - (\alpha + \tfrac{1}{2})(1-p) + \alpha/B, \\
0 &= np - \mu + p/2 + [-\mu - (\alpha + \tfrac{1}{2})(1-p)]x + j(\alpha - n) + yK_c.
\end{aligned}$$

They are exact rational identities in the printed source dictionary.

The retained positive term in (14) is explicitly

$$\begin{aligned}
\mathcal{P} &= m_p(1+u^2)N^2 + (h_0+\alpha)(uN)^2 + \tfrac{1}{2}(vN)^2 + n[v(N + pb_{\mathrm{c}})]^2 \\
&\quad + (C + D + \tfrac{1}{2})(uM)^2 + (C + A + \alpha + \tfrac{1}{2})(vM)^2 + C(uvM)^2 \\
&\quad + d_c(VQ_*)^2 + \tfrac{1}{2}p^2 v^2 b_{\mathrm{c}}^2 + g_c[V(w - UVZ_*)]^2 + \kappa(VQ_* - R_*)^2 + g_c(Q_*^2 + R_*^2).
\end{aligned} \tag{42}$$

In particular $d_c = H^2(A + C + \alpha + \frac{1}{2})y^2 q_z + \kappa > 0$. Combining

$$h_0 N^2 + n(N + jb_{\mathrm{c}})^2 = \widehat{N}^2 + E_*^2, \quad g_c V^2(\Psi - B_*^2) = g_c(\mathcal{E} - \Psi), \quad -\kappa(VQ_* + R_*)^2 = \kappa(VQ_* - R_*)^2 - 2\kappa V^2 Q_*^2 - 2\kappa R_*^2$$

with (41) gives precisely (14) and the complete cross terms (12). Thus the decomposition is an identity for all four state coordinates.

The two radial sums of squares

The radial polynomials used in (30) are

$$\begin{aligned}\mathcal{W} &= 4\Psi + 4U^2Q_*^2 + 4U^2R_*^2 - V^2(U^2-1)^2w^2,\\ \mathcal{W}^\pi &= 4\Psi + 4U^2Q_*^2 + 4U^2B_*^2 - Z_*^2(U^2-1)^2.\end{aligned} \tag{43}$$

Here is their complete positive Gram representation on the chart $U, V \geq 0$, $wZ_* \geq 0$. Put

$$t_7 = \begin{pmatrix} Z_* - U^2Vw \\ Vw - U^2Vw \\ Uw - U^2Vw \\ UZ_* - U^2Vw \\ UVw - U^2Vw \\ UVZ_* - U^2Vw \\ U^2w - U^2Vw \end{pmatrix}, \qquad t_3 = \begin{pmatrix} 1-UV \\ V-UV \\ U-UV \end{pmatrix}$$

and

$$t_7^\pi = \begin{pmatrix} Vw - U^2Z_* \\ Z_* - U^2Z_* \\ Uw - U^2Z_* \\ UVw - U^2Z_* \\ UZ_* - U^2Z_* \\ UVZ_* - U^2Z_* \\ U^2w - U^2Z_* \end{pmatrix}, \qquad t_3^\pi = \begin{pmatrix} w - UZ_* \\ Z_* - UZ_* \\ Uw - UZ_* \end{pmatrix}.$$

With the following two fixed rational matrices,

$$G_7 = \begin{pmatrix} 4 & -2 & -3/2 & 0 & -1/2 & 0 & 1/2 \\ -2 & 3 & 0 & -3/2 & 0 & -1 & 2 \\ -3/2 & 0 & 4 & 3/2 & -2 & -3 & 0 \\ 0 & -3/2 & 3/2 & 4 & -3/2 & 0 & -3/2 \\ -1/2 & 0 & -2 & -3/2 & 3 & 2 & -1 \\ 0 & -1 & -3 & 0 & 2 & 8 & -4 \\ 1/2 & 2 & 0 & -3/2 & -1 & -4 & 4 \end{pmatrix}, \qquad G_3 = \begin{pmatrix} 3 & -2 & -2 \\ -2 & 2 & 2 \\ -2 & 2 & 3 \end{pmatrix},$$

the identities are

$$\begin{aligned}\mathcal{W} &= t_7^T G_7 t_7 + 4VwZ_*(1-U)^2 + UwZ_*\, t_3^T G_3 t_3,\\ \mathcal{W}^\pi &= (t_7^\pi)^T G_7 t_7^\pi + 4VwZ_*(1-U)^2 + UV(t_3^\pi)^T G_3 t_3^\pi.\end{aligned} \tag{44}$$

Their leading principal minors are $(4, 8, 101/4, 1049/16, 907/16, 3279/16, 743/16)$ and $(3, 2, 2)$. Both matrices are therefore positive definite. After separating positive source factors, every radical has a fixed testing argument.

For $wZ_* \leq 0$, one instead uses the following direct expansions:

$$\begin{aligned}\mathcal{W} &= (4U^2 + 3V^2 + 4U^4 + 3U^4V^2 + 2U^2V^2)w^2 + (4 + 4U^2 + 8U^2V^2)Z_*^2 - 16UV(1+U^2)wZ_*,\\ \mathcal{W}^\pi &= (4U^2 + 4V^2 + 4U^4 + 4U^2V^2)w^2 + (3 + 2U^2 + 3U^4 + 8U^2V^2)Z_*^2 - 16UV(1+U^2)wZ_*.\end{aligned} \tag{45}$$

Every displayed monomial is a nonnegative coefficient times a square, using the fixed radical $\sqrt{-UVwZ_*}$ for the last terms. Together with the five-factor expansion of Ψ in the main text, these give a direct sum-of-squares identity also on the negative state chart.

The state signs and testing axes

Let K_d^{test} denote the $d = 1$ core in the numerator of (1), put $h = 1 + u^2 + v^2$, $\Delta = h^2 - u^2v^2 > 0$, and set

$$A_0 = \tfrac{1}{2}\begin{pmatrix} h & -uv \\ -uv & h \end{pmatrix}, \qquad B_0 = \tfrac{1}{2}\begin{pmatrix} -u^2v & -uv^2 \\ uv^2 & u^2v \end{pmatrix}.$$

The exact core completion is

$$K_d^{\mathrm{test}}/D_u = \left[\begin{pmatrix} t \\ r \end{pmatrix} + A_0^{-1}B_0\begin{pmatrix} w \\ \zeta \end{pmatrix}\right]^T A_0 \left[\begin{pmatrix} t \\ r \end{pmatrix} + A_0^{-1}B_0\begin{pmatrix} w \\ \zeta \end{pmatrix}\right] + \frac{D_u D_v}{2\Delta}[u^2v^2(w^2+\zeta^2) - 2uvh\,w\zeta]. \tag{46}$$

Since $A_0 \succ 0$, a negative core requires $uvw\zeta > 0$. Each of the nine original source summands is invariant under the gauges

$$(u, v, W, X, Y, Z) \longmapsto (\varepsilon u, \eta v, W, \eta X, \varepsilon Y, \varepsilon\eta Z).$$

Thus a negative core can always be reduced to $u, v > 0, w\zeta > 0$. At $u = 0, v = 0, w = 0$ or $\zeta = 0$, the remainder in (46) is nonnegative. The explicit identities on both state charts extend to these axes with their literal zero radicals, establishing the theorem on the closed cap.

D A fixed testing family and its affine equations

We now turn the preceding squares into a fixed affine system for the cap in Theorem 1. Their monomial supports determine each Gram block. A finite interpolation grid then proves that the resulting cone is closed.

First take $u, v \geq 0$ and use the common state sign to make $w \geq 0$. There are two closed charts, $\sigma w\zeta \geq 0$, with $\sigma = +1, -1$. All functions below are divided by $\sqrt{D_v}$. Their squared arguments are the following fixed polynomials:

$$q_0 = 1, \quad q_1 = uv, \quad q_2 = uw\zeta, \quad q_3 = vw\zeta \quad (\sigma = +1), \qquad q_0 = 1, \quad q_1 = uv, \quad q_4 = -uvw\zeta \quad (\sigma = -1).$$

These weights are nonnegative on their respective charts. All source radicals in a row are coefficients; for example $\sqrt{UV} = x^{-1/4}y^{-1/4}\sqrt{uv}$ and $\sqrt{UwZ_*} = x^{-1/4}q_z^{-1/4}\sqrt{uw\zeta}$. Thus no source parameter occurs in a testing-function argument.

A finite support prescription

For a row f with just one testing radical $\sqrt{q_k}$, let $\mathrm{supp}_k f$ be the monomials in u, v, t, r, w, ζ that remain after removing that radical and expanding. In taking this support, source-only coefficients are formal symbols: possible specializations to zero never delete a monomial. This convention makes the family independent of the source, including at $n = 0$.

Here is a complete finite prescription for the rows whose supports are used. The ten noncircuit rows of f in the main text are included, as are the three circuit factors on the positive chart or the five factors

$$Uw, \quad Vw, \quad Z_*, \quad UVZ_*, \quad 2\sqrt{-UVwZ_*}$$

on the negative chart. The latter identity is $\Psi = U^2w^2 + V^2w^2 + Z_*^2 + U^2V^2Z_*^2 - 4UVwZ_*$. The retained squares in Appendix C have row supports

$$N,\ uN,\ vN,\ v(N + pb_{\mathrm{c}}),\ uM,\ vM,\ uvM, \quad VQ_*,\ VB_*,\ V(w - UVZ_*),\ VQ_* - R_*,\ Q_*,\ R_*.$$

Every coefficient of these squares is nonnegative on the source domain.

Let $\mathbf{A}, \mathbf{B}, \mathbf{E}$ be the displayed square-root vectors of $\mathcal{A}, \mathcal{B}, \mathcal{E}$ in (23)–(24); let $\mathbf{B}_0 = (R_*, UQ_*, U(w - UVZ_*))$. Let $\mathbf{W}_\sigma, \mathbf{W}_\sigma^\pi$ be the vectors in (44) and (45); on the positive chart their constant positive blocks are $G_7, 1, G_3$. Include every row of

$$\mathbf{W}_\sigma, \quad U\mathbf{A}, \quad U\mathbf{W}_\sigma^\pi, \quad (U^2-1)(UZ_* - B_*), \quad (\alpha/B)R_u - \alpha uR, \quad \mathbf{B}_0,$$
$$v\mathbf{A}, \quad u^2v\mathbf{A}, \quad v\mathbf{B}, \quad u^2v\mathbf{B}, \quad \sqrt{T}\,v(\alpha + c_0u^2)R, \quad \mathbf{A}, \quad \mathbf{B}, \quad v\mathbf{E}, \quad u\mathbf{E}, \qquad c_0 = BTCH.$$

This list contains every positive remainder of the radial, J_α, O, signed-mass and helper identities. In particular, $\bar{a}$ is split into its v^2 and u^4v^2 terms before taking roots; no $\sqrt{\alpha^2 + c_0^2u^4}$ is introduced.

The eight event coordinates are represented by the following rows, with the squared rows in each group summed:

$$\begin{array}{c|l} \lambda & W + vX \\ n & X,\ vX \\ m & vw - t,\ u(vw - t) \\ C & r,\ ur,\ vr,\ uvr \\ A & Y,\ vY \\ D & r + v\zeta,\ u(r + v\zeta) \\ \alpha & vY - Z \\ \beta & t,\ ut,\ vt,\ uvt \end{array}$$

Here W, X, Y, Z are the testing polynomials in (2). These rows allow arbitrary nonnegative event additions literally.

For the four residual pivots take the rows of $\Psi, Q_*^2, b_c^2, R_*^2$, respectively. Their nonconstant loss supports are

$$\begin{array}{c|l} \Psi & (0,2),(0,4),(2,0),(2,2),(4,0),(4,2),(6,2) \\ Q & (0,2),(0,4),(2,0),(4,0),(4,4) \\ E & (0,4),(2,0),(2,2),(4,0),(6,2) \\ R & (0,2),(2,0),(4,2). \end{array} \qquad (*)$$

For each pivot row f and each (a, b) in its row of $(*)$, include $u^{a/2}v^{b/2}f$ in the base supports. Also form a separate block with rows $\sqrt{1 - P_{ab}}\, f$, where $P_{ab} = ((256/9)u)^a(16v)^b$. These rows implement both coefficient signs in the residual identity in the main text. Constant zero coefficients may be retained.

Finally the two signs of the three matrix gates have respective polynomial supports

$$\begin{array}{c|c|l} (a,b) & \text{weight} & \text{monomial vector} \\ (1,1) & 1 \pm P_{11} & (t, r, vw, u\zeta, v\zeta, uw) \\ (0,2),(2,2) & 1 \pm P_{ab} & (r, v\zeta, uw). \end{array}$$

All rows participating in each nonconstant core coefficient lie in the indicated span; the circuit rows do not participate.

For each chart and each radical class, collect the union of its base supports into one monomial vector. For each scalar gate collect the supports belonging to that exponent and class into a separate vector. The matrix gates are the six vectors just displayed. A block with vector m_b and weight q_b represents

$$D_v^{-1} q_b\, m_b^T Q_b m_b, \qquad Q_b \succeq 0.$$

This support prescription is an explicit finite construction, using only polynomial expansion and collection of distinct monomials. Its evaluated vectors and rational weights are in `extraction/fixed_family.json`; the full row list is in `extraction/root_catalog.json`. They give

chart	base	scalar gates	matrix gates	sum
$+$	63	56	24	143
$-$	58	49	24	131.

There are 49 Gram blocks, 2269 upper-triangular Gram variables, and total positive-semidefinite order $143+131+9 = 283$, including the nine scalar source-sign blocks. A constant positive Gram block such as G_7 is transferred to its monomial vector by a congruence with source-only coefficients. Sums of such contributions remain positive. Thus grouping rows does not change any identity or require a mixed inverse.

The rational affine system and its scope

Let $T_c = D_v S_c$ after (2). Its nine coefficient columns are

$$T_\lambda = (W + vX)^2, \quad T_n = D_v X^2, \quad T_m = D_u(vw - t)^2, \quad T_C = D_u D_v r^2, \quad T_A = D_v Y^2,$$
$$T_D = D_u(r + v\zeta)^2, \quad T_\alpha = (vY - Z)^2, \quad T_\beta = D_u D_v t^2, \qquad T_d = (T_D + T_\alpha + T_\beta - D_v Z^2)/2.$$

Let $\mathcal{B}_\sigma$ be the blocks on chart σ. For each chart impose the polynomial coefficient equations

$$\sum_{b \in \mathcal{B}_\sigma} q_b m_b^T Q_b m_b = \sum_{i=1}^{9} c_i T_i.$$

The file `extraction/affine_system.json` records these 554 equations (277 per chart): 2707 Gram coefficients and 132 source coefficients. Indices start at zero; off-diagonal entries are actual symmetric Gram entries and hence contribute twice. Every record is rational.

These equations and positive-semidefinite constraints define a cone K_{283} in the nine source coefficients. The displayed identities and coefficient reserves prove that it contains every native source in (3), including $n = 0$, and the eight event rays. Conversely, each of its sources has $S_c \geq 0$ for all states on the closed testing cap: all block weights are nonnegative there. To cover signed angles, apply

$$(u, v, t, r, w, \zeta) \longmapsto (au, bv, t, abr, bw, a\zeta), \qquad a, b \in \{-1, 1\}.$$

All nine T_i are invariant. Reducing the angles to nonnegative values and then fixing the common state sign reaches one of the two charts. The same formulas include the testing axes and $w\zeta = 0$.

A fixed positive grid proves closedness

Every monomial in m_b has individual degree at most $(4, 3, 1, 1, 1, 1)$ in (u, v, t, r, w, ζ). On either chart choose the 320 rational points in

$$\left\{ \frac{iu_0}{6} : 1 \leq i \leq 5 \right\} \times \left\{ \frac{kv_0}{5} : 1 \leq k \leq 4 \right\} \times \{1/3, 2/3\}^3 \times \sigma\{1/3, 2/3\}.$$

All weights q_b are strictly positive at these points. Distinct monomials with the displayed individual degree bounds are independent on this Cartesian grid by successive univariate interpolation. Consequently the rational matrix

$$H_b = \sum_{a \text{ in grid}} \frac{q_b(a)}{1 + v(a)^2}\, m_b(a) m_b(a)^T$$

is positive definite for every block.

Suppose $c^{(k)} \in K_{283}$ converges to c. Evaluate its identity on these finitely many points and sum. Each summand from each Gram block is nonnegative, so $\operatorname{tr}(H_b Q_b^{(k)})$ is bounded above by the sum of the target values. These target sums converge because they depend linearly on $c^{(k)}$. Since $H_b \succ 0$, every $Q_b^{(k)}$ is bounded. There are finitely many blocks, so a common subsequence converges to positive-semidefinite matrices satisfying the same coefficient equations at c. Hence K_{283} is closed. This argument also handles dependencies between different blocks; it needs only independence within each explicitly chosen monomial vector.

E The complementary Bob angles

For Bob testing angles outside the contact cap, two positive source coefficients control the negative state square directly. This gives a second fixed certificate on $|v| \geq 1/16$, for every Alice angle.

Use the compact testing coordinates

$$(a, b) = \frac{(u, 1)}{\sqrt{1 + u^2}}, \qquad (c, e) = \frac{(v, 1)}{\sqrt{1 + v^2}}, \qquad a^2 + b^2 = c^2 + e^2 = 1.$$

They include the projective endpoints by continuity. In the original state coordinates of (1), put

$$E_B = aW - bY, \quad K_B = aX - bZ, \quad J_B = eE_B + cK_B, \quad U_B = eZ - cY,$$
$$E_\lambda = eW + cX, \qquad L_A = bX + aZ, \qquad M_B = b(cW - eX) + a(cY - eZ).$$

Then the Bell slack has the compact expression

$$\begin{aligned} S_c = {} & \lambda E_\lambda^2 + \beta L_A^2 + nX^2 + mM_B^2 + CE_B^2 + AY^2 + (D + d/2) J_B^2 \\ & + (\alpha + d/2) U_B^2 + \tfrac{d}{2}(L_A^2 - Z^2). \end{aligned} \tag{47}$$

The identity $L_A^2 - Z^2 = X^2 - K_B^2$ leaves one negative square to absorb. Because $J_B - eE_B = cK_B$, the terms with coefficients C and D give exactly

$$\begin{aligned} CE_B^2 + DJ_B^2 - \tfrac{d}{2} K_B^2 = {} & C(cE_B)^2 + (eE_B, J_B) S_B (eE_B, J_B)^T \\ & + \tau_B (c^2 - 1/257) K_B^2, \qquad S_B = \begin{pmatrix} C - \tau_B & \tau_B \\ \tau_B & D - \tau_B \end{pmatrix}, \quad \tau_B = 257d/2. \end{aligned} \tag{48}$$

The last square has nonnegative weight precisely when $|v| \geq 1/16$. The remaining requirement is $S_B \succeq 0$.

For native sources, the estimates in Appendix A give

$$C \geq \frac{f_0}{4R_0HT}, \qquad D \geq \frac{1-\delta^2}{4R_0T(1-H)}.$$

Since $f_0 < 1-\delta^2$, their reciprocal sum is at most $4R_0T[H+(1-H)]/f_0$. Thus

$$\frac{CD}{C+D} \geq \frac{f_0}{4R_0T} > \frac{1}{5T} \geq \frac{1024}{5} > \frac{257}{2}. \tag{49}$$

Here $f_0/(4R_0) = 140600049270784/564050540888577 > 1/5$. The left side of (49) is below both C and D. Since $d = 1$ on the native family, S_B has positive diagonal entries and determinant $CD - \tau_B(C+D) > 0$.

Equations (47)–(48) give the ten fixed functions

$$\Phi_B = (E_\lambda, L_A, X, M_B, Y, U_B, J_B, cE_B, eE_B, \sqrt{c^2 - 1/257}\, K_B)^T.$$

Their matrix $Q_B(c)$ has diagonal

$$(\lambda, \beta, n + d/2, m, A, \alpha + d/2, D - 128d, C, C - 257d/2, 257d/2)$$

and the single off-diagonal pair $(Q_B)_{6,8} = (Q_B)_{8,6} = 257d/2$, with indices starting at zero. It satisfies $S_c = \Phi_B^T Q_B(c) \Phi_B$. Its $(8,6)$ principal block is S_B plus $d/2$ on row 6, and all other entries are nonnegative diagonal terms. At $d = 0$ the same matrix admits each event ray. The formulas therefore include $n = 0$, event additions and finite conic limits. They also apply when $b = 0$ or $e = 0$, covering the projective endpoints in this testing region.

Impose $Q_B(c) \succeq 0$ together with the cap constraints of Appendix D. The resulting system has total order $274 + 10 + 9 = 293$, with the nine source-sign blocks shared. It admits every native source and proves positivity whenever $|u| \leq 9/256$ or $|v| \geq 1/16$. The unaddressed R1 testing region is $|u| > 9/256$, $|v| < 1/16$, including Alice's projective endpoint in this strip. The joined cone is closed: the cap cone is closed, and $Q_B(c)$ is an affine matrix in the source coordinates with no additional variables.

The fifteen upper-triangular rational records of Q_B are in `bob-complement/bob_complement_order10.coo.json`; the adjoining checker verifies every source coefficient in the polynomial identity. `joined_certificate.json` specifies how this matrix and the cap equations share their source coordinates.